

Isocarboxazid

Therapeutics

Brands

- Marplan
- *see index for additional brand names*

Generic?

Not in USA

Class

- Neuroscience-based Nomenclature: serotonin, norepinephrine, dopamine enzyme inhibitor (SN-EI)
- Monoamine oxidase inhibitor (MAOI)

Commonly Prescribed for

(bold for FDA approved)

- **Depression**
- Treatment-resistant depression
- Treatment-resistant panic disorder
- Treatment-resistant social anxiety disorder

How the Drug Works

- Irreversibly blocks monoamine oxidase (MAO) from breaking down norepinephrine, serotonin, and dopamine
- This boosts noradrenergic, serotonergic, and dopaminergic neurotransmission

How Long Until It Works

- Onset of therapeutic actions usually not immediate, but often delayed 2–4 weeks following adequate dosing
- If it is not working within 6–8 weeks, it may require a dosage increase or it may not work at all
- May continue to work for many years to prevent relapse of symptoms

If It Works

- The goal of treatment is complete remission of current symptoms as well as prevention of future relapses

- Treatment most often reduces or even eliminates symptoms, but not a cure since symptoms can recur after medicine stopped
- Continue treatment until all symptoms are gone (remission)
- Once symptoms are gone, continue treating for 1 year for the first episode of depression
- For second and subsequent episodes of depression, treatment may need to be indefinite
- Use in anxiety disorders may also need to be indefinite

If It Doesn't Work

- Many patients have only a partial response where some symptoms are improved but others persist (especially insomnia, fatigue, and problems concentrating)
- Other patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Some patients who have an initial response may relapse even though they continue treatment, sometimes called “poop-out”
- Consider increasing dose, switching to another agent, or adding an appropriate augmenting agent
- Consider psychotherapy
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)
- Some patients may experience apparent lack of consistent efficacy due to activation of latent or underlying bipolar disorder, and require antidepressant discontinuation and a switch to a mood stabilizer

Best Augmenting Combos for Partial Response or Treatment Resistance

- * Augmentation of MAOIs has not been systematically studied, and this is something for the expert, to be done with caution and with careful monitoring
- * A stimulant such as d-amphetamine or methylphenidate (with caution; may activate bipolar disorder and suicidal ideation; may elevate blood pressure)
- Lithium
- Mood-stabilizing anticonvulsants
- Atypical antipsychotics (with special caution for those agents with monoamine reuptake blocking properties, such as lurasidone, ziprasidone, and zotepine)

Tests

- Patients should be monitored for changes in blood pressure; check before and 45–60 minutes after dosing until stable
- For patients receiving high doses or long-term treatment, consider periodic evaluation of hepatic function

* Since MAOIs are frequently associated with weight gain, before starting treatment, weigh all patients and determine if the patient is already overweight (BMI 25.0–29.9) or obese (BMI >30)

- Before giving a drug that can cause weight gain to an overweight or obese patient, consider determining whether the patient already has pre-diabetes (fasting plasma glucose 100–125 mg/dL), diabetes (fasting plasma glucose >126 mg/dL), or dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol), and treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management

* Monitor weight and BMI during treatment

* While giving a drug to a patient who has gained >5% of initial weight, consider evaluating for the presence of pre-diabetes, diabetes, or dyslipidemia, or consider switching to a different antidepressant

Side Effects

How Drug Causes Side Effects

- Theoretically due to increases in monoamines in parts of the brain and body and at receptors other than those that cause therapeutic actions (e.g., unwanted actions of serotonin in sleep centers causing insomnia, unwanted actions of norepinephrine on vascular smooth muscle causing hypotension, etc.)
- Side effects are generally immediate, but immediate side effects often disappear in time

Notable Side Effects

- Dizziness, sedation, headache, sleep disturbances, fatigue, tremor
- Change in appetite, weight gain
- Sexual dysfunction
- Orthostatic hypotension (dose-related); syncope may develop at high doses

Life-Threatening or Dangerous Side Effects

- Hypertensive crisis (especially when MAOIs are used with certain tyramine-containing foods or prohibited drugs)
- Induction of mania in patients with bipolar disorder
- Rare seizures
- Rare hepatotoxicity



Weight Gain

- Many experience and/or can be significant in amount



Sedation

- Many experience and/or can be significant in amount
- Can also cause activation

What to Do About Side Effects

- Wait
- Wait
- Wait
- Lower the dose
- Take at night if daytime sedation
- Switch after appropriate washout to an SSRI or newer antidepressant

Best Augmenting Agents for Side Effects

- Trazodone (with caution) for insomnia
- Benzodiazepines for insomnia

Dosing And Use

Usual Dosage Range

- 40–60 mg/day

Dosage Forms

- Tablet 10 mg

How to Dose

- Initial 10 mg twice a day; increase by 10 mg/day every 2–4 days; dosed 2–4 times/day; maximum dose 60 mg/day



Dosing Tips

- Orthostatic hypotension, especially at higher doses, may require splitting into 3 or 4 daily doses
- Patients receiving high doses may need to be evaluated periodically for effects on the liver
- Little evidence to support efficacy of isocarboxazid at doses below 30 mg/day

Overdose

- Dizziness, sedation, ataxia, headache, insomnia, restlessness, anxiety, irritability; cardiovascular effects, confusion, respiratory depression, or coma may also occur

Long-Term Use

- Consider periodic evaluation of hepatic function
- MAOIs may lose some efficacy long-term

Habit Forming

- Dependence to MAOIs reported but rare

How to Stop

- Generally no need to taper, as drug wears off slowly over 2–3 weeks

Pharmacokinetics

- Clinical duration of action may be up to 14 days due to irreversible enzyme inhibition



Drug Interactions

- Tramadol (serotonin reuptake inhibitor) may increase the risk of seizures in patients taking an MAOI
- Can cause a fatal “serotonin syndrome” when combined with drugs that block serotonin reuptake, so do not use with a serotonin reuptake inhibitor or for 5 half-lives after stopping the serotonin reuptake inhibitor (see [Table 1](#) after Pearls)
- Hypertensive crisis with headache, intracranial bleeding, and death may result from combining MAOIs with sympathomimetic drugs (e.g., amphetamines, methylphenidate, cocaine, dopamine, epinephrine, norepinephrine)
- Do not combine with another MAOI, alcohol, or guanethidine
- Adverse drug reactions can result from combining MAOIs with tricyclic/tetracyclic antidepressants and related compounds, including carbamazepine, cyclobenzaprine, and mirtazapine, and should be avoided except by experts
- MAOIs in combination with spinal anesthesia may cause combined hypotensive effects
- Combination of MAOIs and CNS depressants may enhance sedation and hypotension



Other Warnings/Precautions

- Use requires strict adherence to low-tyramine diet (see [Table 2](#) after Pearls)
- Patient and prescriber must be vigilant to potential interactions with any drug, including antihypertensives and over-the-counter cough/cold preparations

- Over-the-counter medications to avoid include cough and cold preparations, including those containing dextromethorphan (serotonin reuptake inhibitor), nasal decongestants (tablets, drops, or spray), hay-fever medications, sinus medications, asthma inhalant medications, anti-appetite medications, weight reducing preparations, “pep” pills (see [Table 3](#) after Pearls)
- Hypoglycemia may occur in diabetic patients receiving insulin or oral antidiabetic agents (hydrazine effect)
- Binds and inactivates vitamin B6 (hydrazine effect); reversible with high-dose B6 administration
- Use cautiously in patients receiving reserpine, anesthetics, disulfiram, metrizamide, anticholinergic agents
- Isocarboxazid is not recommended for use in patients who cannot be monitored closely

Do Not Use

- If patient is taking meperidine (pethidine) (serotonin reuptake inhibitor)
- If patient is taking a sympathomimetic agent or taking guanethidine
- If patient is taking another MAOI
- If patient is taking any agent that can inhibit serotonin reuptake (e.g., SSRIs, tramadol, milnacipran, duloxetine, venlafaxine, clomipramine, imipramine, etc.)
- If patient is taking dextromethorphan, a serotonin reuptake inhibitor
- If patient is taking diuretics or other antihypertensives without careful monitoring of blood pressure
- If patient has pheochromocytoma
- If patient has cardiovascular or cerebrovascular disease, until cleared by the patient’s medical doctor
- If patient is taking a prohibited drug
- If patient is not compliant with a low-tyramine diet
- If there is a proven allergy to isocarboxazid

Special Populations

Renal Impairment

- Use with caution –drug may accumulate in plasma
- May require lower than usual adult dose

Hepatic Impairment

- Use cautiously in hepatic impairment

Cardiac Impairment

- Contraindicated in patients with congestive heart failure or hypertension, unless cleared by the patient's medical doctor

Elderly

- Initial dose lower than usual adult dose
- Elderly patients may have greater sensitivity to adverse effects, but many tolerate MAOIs
- Reduction in the risk of suicidality with antidepressants compared to placebo in adults age 65 and older

Children and Adolescents

- Not recommended for use in children under age 16
- Carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient's chart
- Distribute the brochures provided by the FDA and the drug companies
- Whenever possible, warn patients and their caregivers about the possibility of activating side effects, and advise them to report such symptoms immediately
- Use with caution, observing for activation of known or unknown bipolar disorder and/or suicidal ideation, and inform parents or guardians of this risk so they can help observe child or adolescent patients

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Not generally recommended for use during pregnancy, especially during first trimester
- Should evaluate patient for treatment with an antidepressant with a better risk/benefit ratio
- National Pregnancy Registry for Antidepressants: 1-866-961-2388, <https://womensmentalhealth.org/research/pregnancyregistry/antidepressants>

Breast Feeding

- Some drug is found in mother's breast milk
- Immediate postpartum period is a high-risk time for depression, especially in women who have had prior depressive episodes, so drug may need to be reinstituted late in the third trimester or shortly after childbirth to prevent a recurrence during the postpartum period
- Should evaluate patient for treatment with an antidepressant with a better risk/benefit ratio

The Art Of Psychopharmacology

Potential Advantages

- Atypical depression
- Severe depression
- Treatment-resistant depression or anxiety disorders

Potential Disadvantages

- Requires compliance to dietary restrictions, concomitant drug restrictions
- Patients with cardiac problems or hypertension, unless cleared by the patient's medical doctor
- Multiple daily doses, although many patients can be dosed once a day

Primary Target Symptoms

- Depressed mood
- Somatic symptoms
- Sleep and eating disturbances
- Psychomotor retardation
- Morbid preoccupation

Pearls

- MAOIs are generally reserved for third- or fourth-line use after SSRIs, SNRIs, and combinations of newer antidepressants have failed
- Despite little utilization, some patients respond to isocarboxazid who do not respond to other antidepressants including other MAOIs
- Patient should be advised not to take any prescription or over-the-counter drugs without consulting their doctor because of possible drug interactions with the MAOI
- Headache is often the first symptom of hypertensive crisis
- The rigid dietary restrictions may reduce compliance (see [Table 2](#) after Pearls)
- Mood disorders can be associated with eating disorders (especially in adolescent females), and isocarboxazid can be used to treat both depression and bulimia
- * Myths about the danger of dietary tyramine can be exaggerated, but prohibitions against concomitant drugs often not followed closely enough
- Orthostatic hypotension, insomnia, and sexual dysfunction are often the most troublesome common side effects

* MAOIs should be for the expert, especially if combining with agents of potential risk (e.g., stimulants, trazodone, TCAs)

* MAOIs should not be neglected as therapeutic agents for the treatment-resistant

- For treatment-resistant patients, experts can give a tricyclic/tetracyclic antidepressant other than clomipramine or imipramine simultaneously with an MAOI for patients who fail to respond to numerous other antidepressants
- Use of MAOIs with clomipramine is always prohibited because of the risk of serotonin syndrome and death due to serotonin reuptake inhibition
- Amoxapine may be the preferred tricyclic/tetracyclic antidepressant to combine with an MAOI in heroic cases due to its theoretically protective 5HT_{2A} antagonist properties
- If this option is elected, start the MAOI with the tricyclic/tetracyclic antidepressant simultaneously at low doses after appropriate drug washout, then alternately increase doses of these agents every few days to a week as tolerated
- Although very strict dietary and concomitant drug restrictions must be observed to prevent hypertensive crises and serotonin syndrome, the most common side effects of MAOI and tricyclic/tetracyclic combinations may be weight gain and orthostatic hypotension

Table 1. Drugs contraindicated due to risk of serotonin syndrome/toxicity

Do Not Use:			
Antidepressants	Drugs of Abuse	Opioids	Other
SSRIs	MDMA (ecstasy)	Meperidine	Non-subcutaneous sumatriptan
SNRIs	Cocaine	Tramadol	Chlorpheniramine
Clomipramine	Methamphetamine	Methadone	Brompheniramine
Imipramine	St. John's wort	High-dose or injected amphetamine	Dextromethorphan Lumateperone Ziprasidone

Table 2. Dietary guidelines for patients taking MAOIs*

Foods to avoid**	Foods allowed
Dried, aged, smoked, fermented, spoiled, or improperly stored meat, poultry, and fish	Fresh or processed meat, poultry, and fish; properly stored pickled or smoked fish
Broad bean pods	All other vegetables
Aged cheeses	Processed cheese slices, cottage cheese, ricotta cheese, yogurt, cream cheese

Foods to avoid**	Foods allowed
Tap and unpasteurized beer	Canned or bottled beer and alcohol
Marmite	Brewer's and baker's yeast
Banana peel	Bananas, avocados, raspberries
Sauerkraut, kimchee	Peanuts
Soy products/tofu	
Tyramine-containing nutritional supplement	

* Consult up-to-date listing of tyramine content, as most foods now have low tyramine amounts**Not necessary for 6 mg transdermal or low-dose oral selegiline

Table 3. Drugs that boost norepinephrine: should only be used with caution with MAOIs

Use With Caution:			
Decongestants	Stimulants	Antidepressants with norepinephrine reuptake inhibition	Other
Phenylephrine	Amphetamines	Most tricyclics	Phenylephrine
Pseudoephedrine	MethylphenidateCocaineMethamphetamineModafinilArmodafinil	NRIsNDRIs	Local anesthetics containing vasoconstrictors

Suggested Reading

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Kim T, Xu C, Amsterdam JD. Relative effectiveness of tricyclic antidepressant versus monoamine oxidase inhibitor monotherapy for treatment-resistant depression. *J Affect Disord* 2019 ;250:199–203.

Ketamine

Therapeutics

Brands

- Ketalar
- *see index for additional brand names*

Generic?

Yes

Class

- N-methyl-D-aspartate (NMDA) receptor antagonist

Commonly Prescribed for

(bold for FDA approved)

- **Induction and maintenance of general anesthesia**
- Pain/neuropathic pain
- Sedation
- Treatment-resistant depression (experimental)

How the Drug Works

- Ketamine is a noncompetitive open channel inhibitor of the NMDA receptor; specifically, it binds to the phencyclidine site of the NMDA receptor
- This leads to downstream glutamate release and consequent stimulation of other glutamate receptors, including AMPA receptors
- Theoretically, ketamine may have antidepressant effects at low (subanesthetic) doses because activation of AMPA receptors leads to activation of signal transduction cascades that cause the expression of synaptic proteins and an increase in the density of dendritic spines
- Low (subanesthetic) doses also produce analgesia and modulate central sensitization, hyperalgesia, and opioid tolerance

How Long Until It Works

- For treatment-resistant depression, antidepressant effects can occur within hours
- For neuropathic pain, effects can occur within hours but may take weeks for full effect

If It Works

- For treatment-resistant depression, can immediately alleviate depressed mood and suicidal ideation, but antidepressant effects last only a few days
- For neuropathic pain, can continue to use as long as it is beneficial

If It Doesn't Work

- Try a traditional antidepressant or electro-convulsive therapy (ECT) for treatment-resistant depression
- Try traditional analgesics and treatments for neuropathic pain



Best Augmenting Combos for Partial Response or Treatment Resistance

- For neuropathic pain, may use cautiously with opioids
- For treatment-resistant depression, combinations have not been systematically studied

Tests

- None for healthy individuals

Side Effects

How Drug Causes Side Effects

- Direct effect on NMDA receptors

Notable Side Effects

- When used as an anesthesia induction/maintenance agent (generally at doses >2 mg/kg IV), it may produce emergent psychosis, including auditory and visual hallucinations, restlessness, disorientation, vivid dreams, and irrational behavior. Spontaneous

involuntary movements, nystagmus, hypertonus, and vocalizations are also common. These adverse effects are uncommon with very low-dose therapy

- CSF pressure increased, erythema (transient), morbilliform rash (transient), anorexia, pain/erythema at the injection site, exanthema at the injection site, skeletal muscle tone enhanced, intraocular pressure increased, bronchial secretions increased, potential for dependence with prolonged use, emergence reactions (includes confusion, dreamlike state, excitement, irrational behavior, vivid imagery)
- Psychotomimetic phenomena (euphoria, dysphasia, blunted affect, psychomotor retardation, vivid dreams, nightmares, impaired attention, memory and judgment, illusions, hallucinations, altered body image), delirium, dizziness, diplopia, blurred vision, nystagmus, altered hearing, hypertension, tachycardia, hypersalivation, nausea and vomiting, erythema and pain at injection site
- Urinary tract toxicity
- When used at higher doses in anesthesia, tonic–clonic movements are very common (>10%); however, these have not been reported after oral use or with the lower parenteral doses used for analgesia



Life-Threatening or Dangerous Side Effects

- Syncope or cardiac arrhythmias
- Hypertension/hypotension

- Anaphylaxis
- CNS depression
- Respiratory depression/apnea
- Airway obstruction/laryngospasm

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount

What to Do About Side Effects

- Pretreatment with a benzodiazepine reduces incidence of psychosis by >50%
- For CNS side effects, discontinuation of nonessential centrally acting medications may help

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- Oral: 10–50 mg
- IV infusion: 1–10 $\mu\text{g/kg}$ per minute

Dosage Forms

- Oral solution: 50 mg/mL
- Injection: 50 mg/mL, 100 mg/mL

How to Dose

- For pain (oral): initial 10 mg; titrate up as appropriate
- For pain (IM): 2–4 mg/kg
- For pain (IV): 0.2–0.075 mg/kg
- For pain (continuous IV infusion): 2–7 $\mu\text{g/kg}$ per minute



Dosing Tips

- Slow titration can reduce side effects
- Food does not affect absorption
- For oral use: to prepare 100 mL of 50 mg/5 mL ketamine oral solution
 - 2 x 10 mL vials of generic ketamine 50 mg/mL for injection (cheapest concentration)
 - 80 mL purified water

- Store in a refrigerator with an expiry date of 1 week from manufacture
- Patients can add their own flavoring, e.g., fruit cordial, just before use to disguise the bitter taste
- For sublingual use:
 - Place under the tongue and ask patient not to swallow for 2 minutes
 - Use a high concentration to minimize dose volume; retaining >2 mL is difficult
 - Start with 10 mg
- Incompatibility
 - Ketamine forms precipitates with barbiturates and diazepam (manufacturer's data on file); do not mix
 - Mixing lorazepam with ketamine is also not recommended; compatibility data are lacking, and there is a risk of adsorption of lorazepam to the tubing

Overdose

- Restlessness, psychosis, hallucinations, stupor, respiratory depression

Long-Term Use

- Long-term cognitive and memory impairment have been reported with repeated ketamine misuse or abuse

Habit Forming

- No

How to Stop

- Taper not necessary

Pharmacokinetics

- Plasma half-life: alpha: 10–15 minutes; beta: 2.5 hours; 1–3 hours IM; 2.5–3 hours orally; 12 hours norketamine
- Metabolized by CYP450 2B6, 2C9, and 3A4



Drug Interactions

- Use with caution with other drugs that are NMDA antagonists (amantadine, memantine, dextromethorphan)
- Ketamine may increase the effects of other sedatives, including benzodiazepines, barbiturates, opioids, anesthetics, and alcohol
- CYP450 3A4 inhibitors (e.g., clarithromycin, ketoconazole) can increase plasma concentrations of ketamine and reduce those of norketamine, but the clinical relevance of this is unclear
- Plasma concentrations of ketamine are increased by diazepam

- Barbiturates and hydroxyzine may increase the effects of ketamine; avoid combination

Other Warnings/Precautions

- Use with caution in patients with current or past history of psychiatric disorder; epilepsy, glaucoma, hypertension, heart failure, ischemic heart disease, and a history of cerebrovascular accidents

Do Not Use

- If patient has a condition in which an increase in blood pressure would be hazardous
- If patient has schizophrenia or another psychotic disorder
- If patient has a condition in which an increase in intraocular pressure would be hazardous
- If there is a proven allergy to ketamine

Special Populations

Renal Impairment

- Reduce dose for moderate impairment
- Should not be used in severe impairment

Hepatic Impairment

- Dose reduction not necessary

Cardiac Impairment

- Use with caution

Elderly

- Some patients may tolerate lower doses better



Children and Adolescents

- Safety and efficacy have not been established



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Use only if potential benefits outweigh the potential risks to the fetus
- National Pregnancy Registry for Antidepressants: 1-866-961-2388, <https://womensmentalhealth.org/research/pregnancyregistry/antidepressants>

Breast Feeding

- Unknown if ketamine is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed

The Art of Psychopharmacology

Potential Advantages

- For pain, may be especially useful when used in conjunction with opioids
- Severely treatment-resistant depression, suicidal ideation

Potential Disadvantages

- May produce dysphoria, nightmares, excitement (uncommon with very low-dose therapy)
- Antidepressant effects are short-lived

Primary Target Symptoms

- Pain
- Treatment-resistant depression

Pearls

- Actions in treatment-resistant depression are transient, lasting only a few days following infusion

- The use of ketamine can cause urinary tract symptoms (e.g., frequency, urgency, urge incontinence, dysuria, and hematuria); the causal agent has not been determined, but direct irritation by ketamine and/or its metabolites is a possibility. (Investigations have revealed interstitial cystitis, detrusor overactivity, decreased bladder capacity; symptoms generally settle several weeks after stopping ketamine.)
- May be used in combination with anticholinergic agents to decrease hypersalivation
- Do not mix with barbiturates or diazepam (precipitation may occur)
- Bronchodilation is beneficial in asthmatic or chronic obstructive pulmonary disease (COPD) patients. Laryngeal reflexes may remain intact or may be obtunded
- The direct myocardial depressant action of ketamine can be seen in stressed, catecholamine-deficient patients
- Ketamine releases endogenous catecholamines (epinephrine, norepinephrine), which maintain blood pressure and heart rate, and increase myocardial oxygen demand
- Ketamine increases cerebral metabolism and cerebral blood flow while producing a noncompetitive block of the neuronal postsynaptic NMDA receptor
- Lowers seizure threshold
- Recent laboratory/clinical studies support the use of low-dose ketamine to improve postoperative analgesia/outcome

- May be especially beneficial for refractory neuropathic pain/complex regional pain syndrome
- May be especially beneficial when used in conjunction with opioids
- (S)-ketamine is available in a preservative-free solution in Europe; however, it currently is not approved by the FDA. S(+)-ketamine may be more potent and have fewer side effects when used intravenously than the racemate. Although not rigorously tested and not available in the USA; some European investigators have utilized the preservation-free solution for intrathecal/epidural use –this is not recommended

Suggested Reading

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Lamotrigine

Therapeutics

Brands

- Lamictal
- Labileno
- Lamictin
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: glutamate, voltage-gated sodium channel blocker (Glu-CB)
- Anticonvulsant, mood stabilizer, voltage-sensitive sodium channel antagonist

Commonly Prescribed for

(bold for FDA approved)

- **Maintenance treatment of bipolar I disorder**